Enhancing Profitability and Cash Flow at a Grocery Store: A Data-Driven Approach to Credit, Financial, and Margin Management

A Final report for the BDM capstone Project

Submitted by

Name: Mohammad Kashan

Roll number: 23F2003821

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1. Executive Summary 2

2. Detailed Explanation of Analysis Process/Method 2

3. Results and Findings 6

4. Interpretation of Results and Recommendation ……………………………17

5. Acknowledgements ………………………………..……….………..…. 21

6. Project Links and Datasets …...…………………...……..……..…………..… 21

1. Executive Summary

Faiz General Store, an established local kirana shop in Prayagraj, faces a critical profitability challenge despite stable revenue. The business is caught in a "Volume vs. Value Trap," operating on a thin 9.45% net margin. This analysis identified two root causes: a deteriorating cash-to-credit ratio (falling from 3.42 to 1.86) and a poor sales mix that neglects high-margin categories.

The project analyzed 12 months of daily financial data, complemented by a 6-month deep dive into category and credit transactions. A proprietary Reliability Score (70% default-weighted) was built to segment all 105 customers. A BCG-style Strategy Matrix was used to analyze the product portfolio, revealing the strategic misalignment.

Key findings were significant: 98% of all credit defaults (Rs 39k) originate from the "Watchlist" and "Cash Only" tiers, with the "Cash Only" tier proved to be unprofitable (-11.8% margin). Furthermore, the store's most profitable categories (Snacks, Personal Care) are stagnant, while peak weekend traffic (22% higher revenue) is not being strategically utilized.

A two-part strategy is recommended: 1) Immediately implement the Reliability Score policy to move all 28 "Cash Only" customers to a Rs 0 limit, stopping the profit leak. 2) Use peak weekend traffic to promote the high-margin "Question Mark" categories. This holistic, data-driven strategy is projected to deliver a 43.4% net profit increase—transforming Faiz General Store from a struggling 9.45% net margin operation into a professionally managed, sustainably profitable 13.5% net margin enterprise.

2. Detailed Explanation of Analysis Process/Method

The analysis was conducted using Python with the Pandas library for data manipulation, Matplotlib/Seaborn for visualization, and Google Colab as the execution environment. The methodology was structured to address each of the three core business problems identified.

2.1 Data Cleaning and Preprocessing

- **Explanation:** Before any analysis, all three raw datasets (Raw-Credit, Raw-Category-wise, Raw-Day-wise) were loaded into Pandas DataFrames. The cleaning process involved:
  1. **Type Conversion:** All date columns were converted to a standard datetime format using pd.to_datetime() to enable time-series analysis. All monetary and numerical columns (e.g., Revenue, Cost, Credit_amt) were converted to numeric types using pd.to_numeric(), with errors coerced into NA.
  2. **Missing Value Imputation:** NA values in numerical columns were filled with 0. This was a deliberate business choice; for example, a missing Amount_repaid implies Rs 0 has been paid so far. For the 48 rows with missing Payment_date in the credit data, a 15-day average delay was imputed, as detailed in the mid-term report. For a total of 5 closing days, the shop was closed because of famous muslim festivals, so people know this advance, so they bought all the items in advance, so filling 0 is perfect for balancing the normalization.
  3. **Categorical Standardization:** In the Payment_status column, text-based inconsistencies (e.g., "part" and "partial") were standardized to a single value ("partial") to ensure accurate grouping.
- **Importance:** This cleaning process was critical for ensuring data quality and analytical accuracy. Without standardizing data types, calculations of sums or averages would fail. Without proper imputation and standardization, any subsequent analysis, especially the Reliability Score, would be based on "garbage in, garbage out" (GIGO), leading to inaccurate findings and flawed business recommendations.

2.2 Comprehensive Explanation for each Method/Analysis Used

The analysis was broken down by problem statements to directly link methods to business goals.

2.2.1 Advanced Credit Management (Reliability Score Model)

- **Explanation:** The mid-term S/A/B/C/D/E/F rating system was a good rule-based start but lacked granularity. To create a truly data-driven and actionable policy, a numerical **Customer Reliability Score** (0-100) was developed. This score is a weighted composite based on the two key predictors of risk: default history and payment behavior.
- **Process:** The foundational analysis for both the mid-term S/A/B/C/D/E/F ratings and the new Reliability Score relied on a central **customer summary table** generated in Python. Using the raw credit data, all transactions were grouped by Customer_name to calculate key aggregate metrics: Total_credit, Total_repaid, Total_defaulted, Default_Percentage, and Average_delay. These same, validated metrics served as the primary inputs for the new, more advanced scoring model.
- **Abstraction (Mathematical Definition):** The score is calculated as a weighted sum of a Default Score (70% weight) and a Delay Score (30% weight).

$$\text{Reliability Score} = \text{Default Score} + \text{Delay Score}$$

  - **Default Score (70 points):** This component is designed to be highly punitive, as defaults are the primary source of financial loss. The formula (1 - (3 × default_percentage/100)) ensures that even a small default rate results in a major score penalty.

    $$\text{Default Score} = 70 \times \max(0, (1 - (3 \times \text{default\_percentage}/100)))$$
  - **Delay Score (30 points):** This component measures payment timeliness. The Average_delay is normalized against a 45-day cap, which was identified as a reasonable "worst-case" delay period from the data.

    $$\text{Delay Score} = 30 \times (1 - \min(\text{avg\_delay}/45, 1))$$
- **Abstraction (Strategic Tiering):** To translate the 0-100 numerical score into an actionable business policy, customers were segmented into four distinct **Customer Tiers** based on clear, data-driven thresholds. These tiers form the basis for all credit recommendations.

1. **Premium:** Reliability Score >= 80
2. **Standard:** Reliability Score 65-79

3. **Watchlist:** Reliability Score 50-64

4. **Cash Only:** Reliability Score < 50

- **Justification:** This two-step model(Score + Tiers) is analytically superior to the rule-based rating's rigid rules. It directly addresses the "concentrated credit risk" problem (shown in Figure 8) by providing a single, precise metric for every customer. This score and its tiers are the engine for our final Limit = Score × 20 policy, allowing for a fair, scalable, and automated solution that the simple S/A/B/C/D/E/F system could not provide.

2.2.2 Category-wise Strategic Analysis (BCG Matrix)

- **Explanation:** To solve the "suboptimal profit margin" problem, we needed to move beyond simple revenue totals and analyze the strategic position of each category. A **BCG-style Strategy Matrix** was created to visualize the "Volume vs. Value Trap."

- **Abstraction (Matrix Definition):** The matrix plots four data dimensions:
  - **X-axis: Average Profit Margin (%)**. The threshold was set at 12.5%, the shop's own data-driven average margin.
  - **Y-axis: Revenue Share (%)**. The threshold was set at 10% to separate high- and low-volume categories.
  - **Bubble Size: Total Revenue (Rs)**. This visualizes the absolute financial impact of each category.
  - **Quadrants:** Categories are segmented into "Stars," "Cash Cows," "Question Marks," or "Dogs" based on their position.
- **Justification:** This method was chosen because it is the standard professional framework for portfolio analysis. It perfectly illustrates the core problem: the Staples category (a "Cash Cow") drives high volume but low profit, while Personal Care and Snacks ("Question Marks") are high-profit but low-volume. This analysis directly guides the final recommendation to rebalance promotions.

2.2.3 Financial Health Trend Analysis

- **Explanation:** To monitor the store's overall health and the severity of the identified problems over time, two Key Performance Indicators (KPIs) were tracked over the full 12-month period.
- **Abstraction (KPI Definitions):**
  1. **Liquidity Trend (Figure 1):** The **Cash-to-Credit Ratio** was calculated monthly. This metric is a direct proxy for cash flow health and credit dependency.
     Ratio = Total Monthly Cash Sales / Total Monthly Credit Sales
  2. **Performance Trend (Figure 3):** The **Total Profit** (in Rs) was calculated monthly to track overall business performance relative to its revenue.
     Profit = Total Monthly Revenue - Total Monthly Cost
- **Justification:** These methods were chosen to provide a high-level diagnostic of the business. The Cash-to-Credit Ratio trend (Figure 1) directly confirms the "deteriorating cash flow" problem, justifying the urgency of the new credit policy. The Profit trend (Figure 3) confirms the "suboptimal profitability" problem, justifying the need for the category analysis.

3. Results and Findings

All the results and findings are presented in a manner to give a complete overview of the situation and analysis.

3.1 Financial Health Analysis:

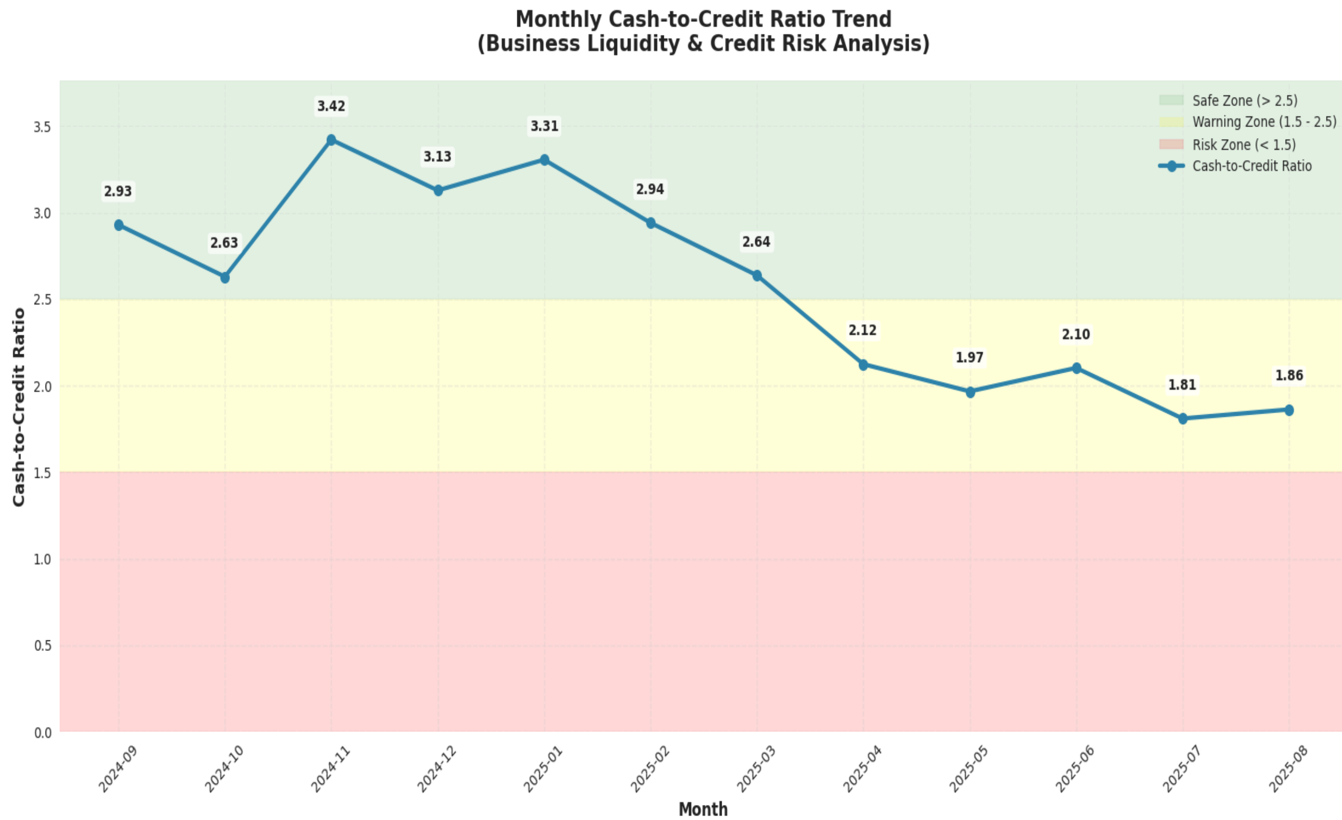

Figure 1: Cash-to-Credit Ratio Trend Analysis

This chart tracks the ratio of cash-to-credit sales over twelve months, revealing a significant downward trend.The ratio line has fallen from safe zone to warning zone, indicating the business is becoming increasingly reliant on credit sales. The ratio has fallen from its peak of 3.42 to current 1.86 with the decline rate of 15.6% per month.This shift represents a critical financial risk, as it negatively impacts the shop's immediate cash flow and increases its exposure to bad debt from non-paying customers. The less cash to credit ratio also shows that now the shop has less ability to promote the business, give discounts and invest in profitable categories because of less in hand cash sales and waiting for credit payment which never paid in full instead in parts. It makes the shop more dependent and lowers its ability to grow.

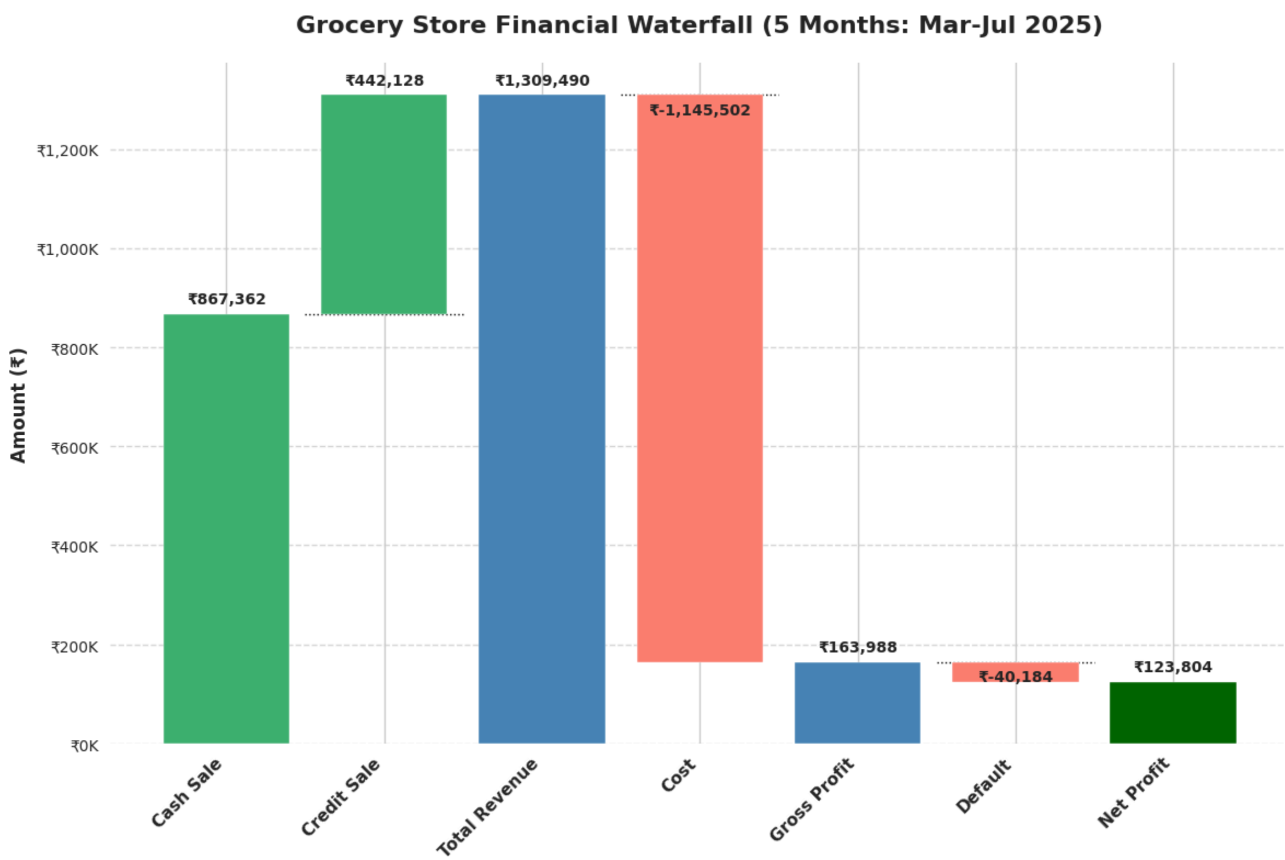

Figure 2: Grocery Store Complete Financial Waterfall Chart

This graph tells us the complete story of the shop's financials, from revenue to net profit. It clearly shows that the shop got Rs 8,67,362 in cash sale, 66% of revenue and Rs 4,42,128 from credit sale, 34% of total revenue. In 5 months total of Rs 13,09,490 revenue earned from which Rs 11,45,502 deducted in cost, after that shop earned Rs 1,63,988 in gross profit, which is 12.5% of the revenue, but after that shop losses Rs 40,184 in default and ended up with a net profit of Rs 1,23,804 which is 9.45% of total revenue. The average monthly income of the shop is Rs 24,760 which is not so good. The shop loses 24% of its gross profit in default, otherwise its average monthly income would be Rs 33k per month. The reason for this thin gross profit margin is poor strategic management and focus on less profitable products, while the reason for this Rs 40,184 default which is 9% of total revenue is poor credit management, no reminder system for credit, no data driven and perfectly written and managed system for credit.

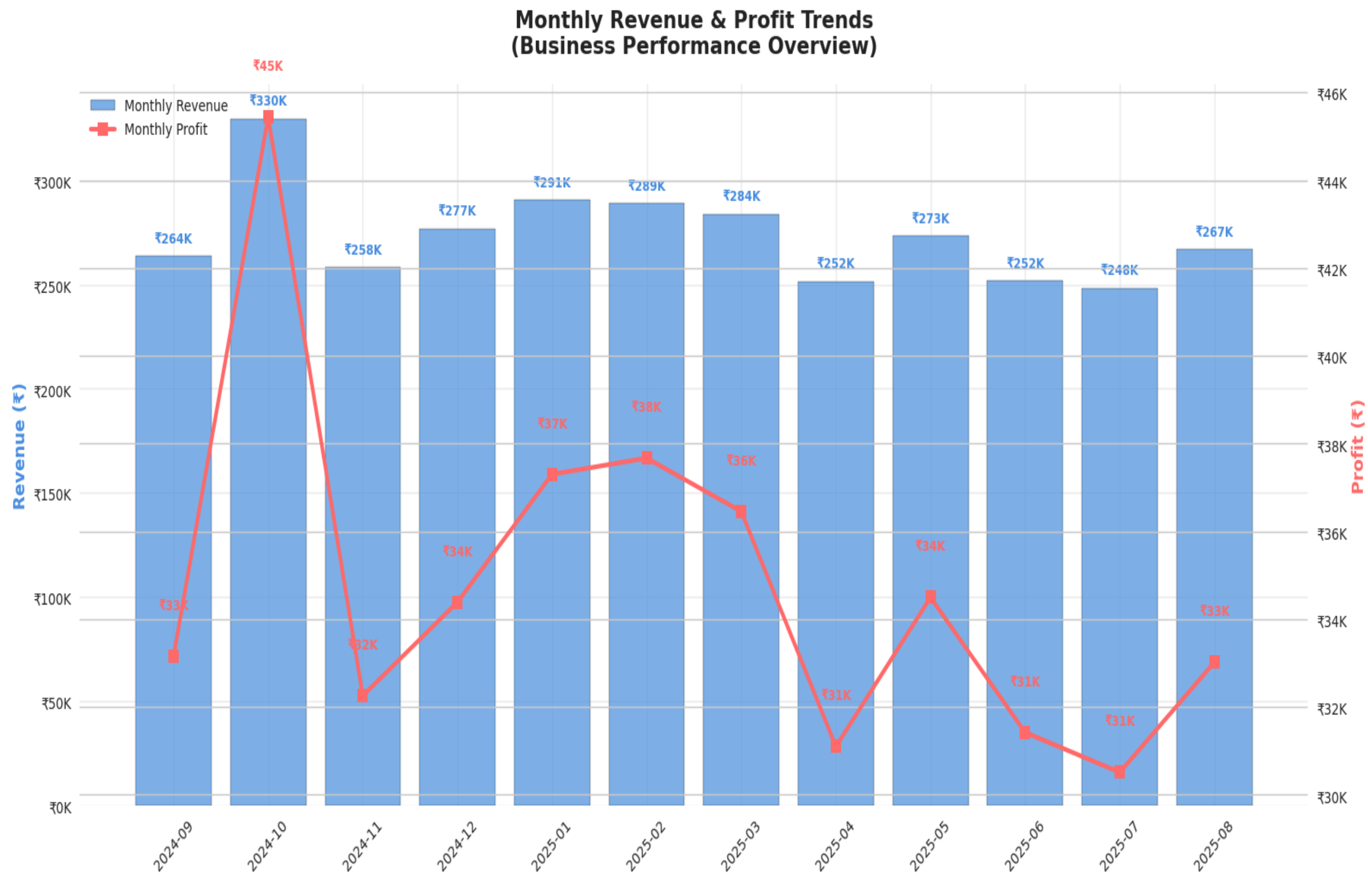

Figure 3: Revenue & Profitability Trend Analysis

This 12-month dual-axis chart shows Total Revenue (bars, left axis) and the resulting Total Profit (line, right axis).

Revenue is relatively stable, indicating a consistent customer base, but also shows the seasonality, as we can see that revenue is highest Rs 330k in October month, which is festival month, also revenue is at good high in Jan 25 Rs 291k, Feb 25 Rs 289k due to the effect of Kumbh Mela, revenue is above average in March 25 Rs 284k also because of 1 month muslim festival Ramzan and in Ramzan Snacks sale is on peak which drives the revenue to above average. However, profit is far more volatile but somehow follows the revenue trend as you can see where revenue is on high profit also good, the volatility suggests that high-revenue months were driven by low-margin products. This reinforces the need for the strategy change, to shift the sales mix toward more profitable items to ensure that sales growth translates into profit growth.

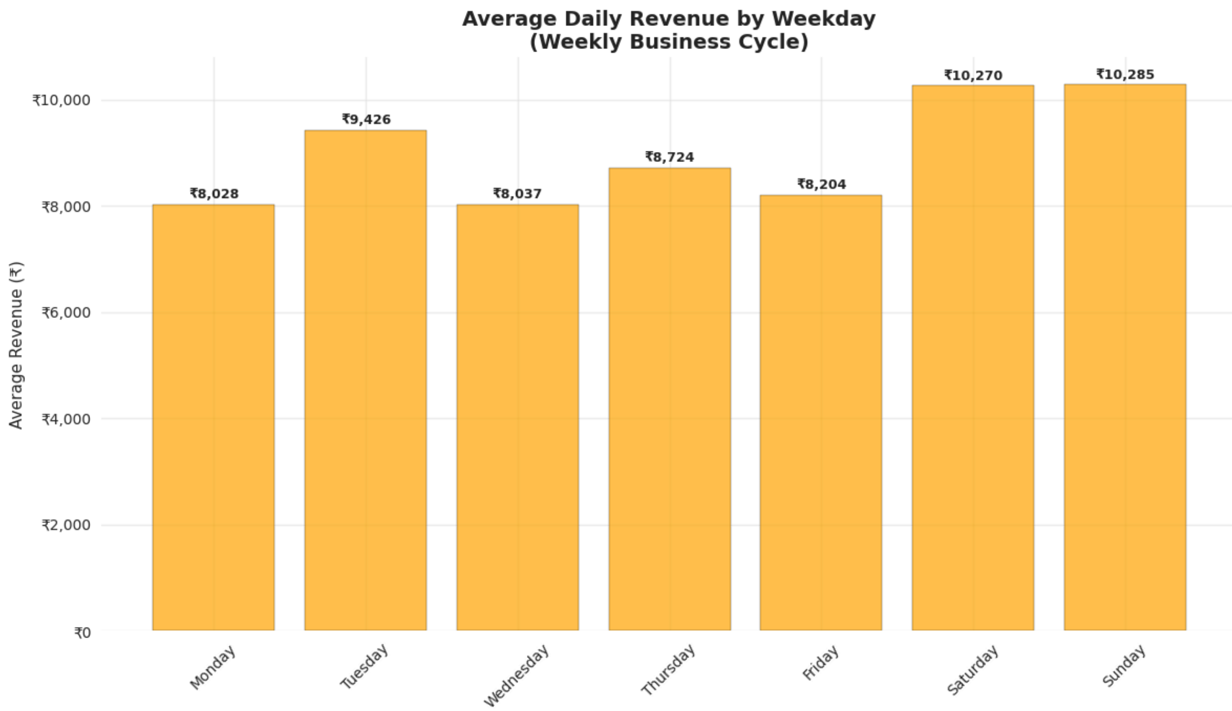

Figure 4: Weekly Revenue Trend Analysis

This bar chart analyzes the weekly business cycle by plotting the average revenue for each day of the week.

A clear and actionable pattern is visible. Revenue is lowest on Mondays and Wednesdays but spikes significantly on the weekend, with Saturday and Sunday being the busiest days by over 22%. The reason for that pattern is that on weekends people get relief from their work so they celebrate it with some beverages or snacks etc. School students also buy items from this shop instead of their school's nearest shop because they are at home on Sunday.

This is a key strategic insight. The weekend represents peak customer traffic. The shop can use this opportunity to concentrate all promotions for high-margin categories (like Snacks and Personal Care) on Saturdays and Sundays. This uses the store's busiest time to solve its biggest profitability problem.

3.2 Category-wise Analysis:

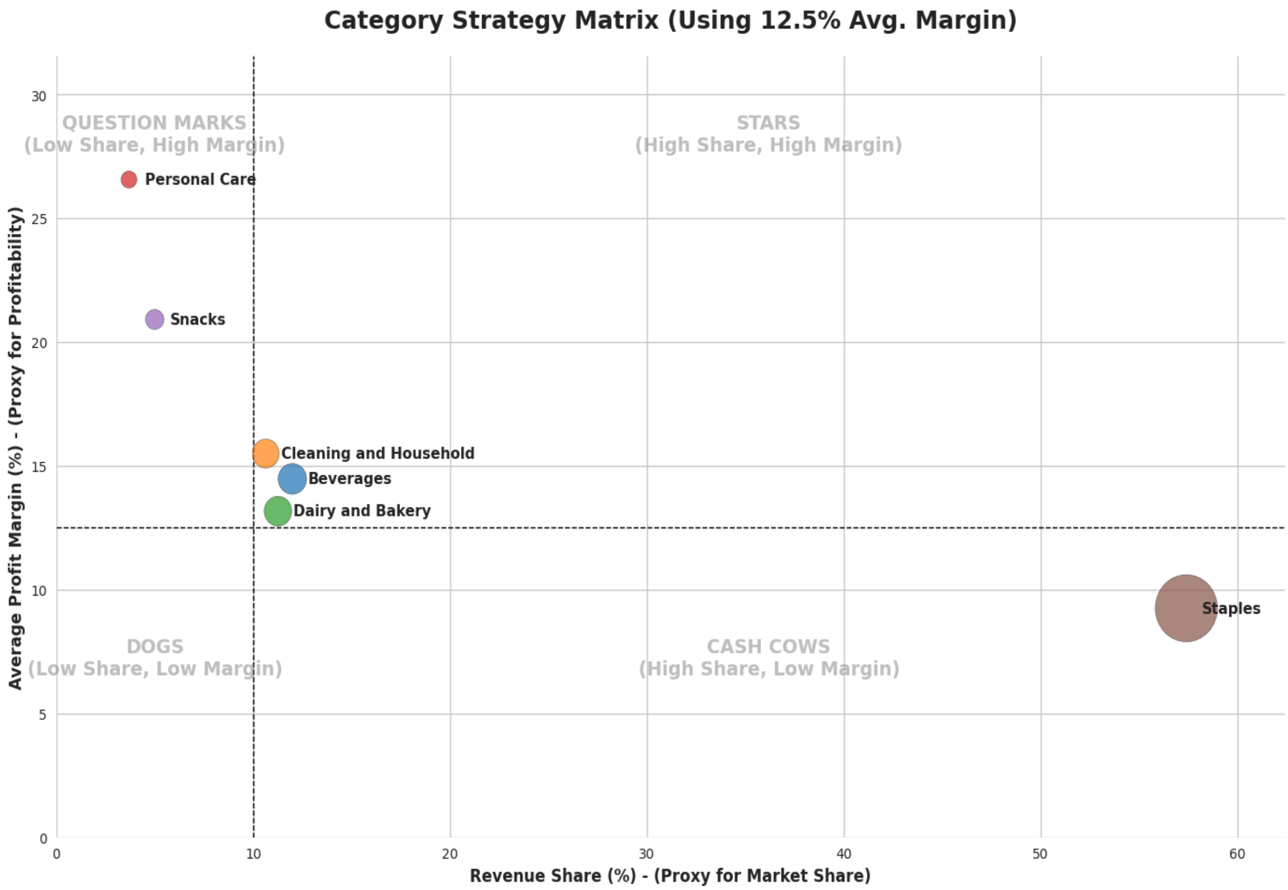

Figure 5: Strategic Category Positioning - BCG Matrix Analysis

This matrix presents a strategic overview of the store's product categories. The 12.5% line represents the shop's average margin, and the 10% line separates high/low revenue share. The bubble size is proportional to the total revenue (Rs) contributed by that category.

The matrix reveals a "Profitability Paradox." The largest bubble, Staples, is a Cash Cow: it drives the majority of revenue (57%) but operates at a low 9.25% margin. Conversely, Personal Care and Snacks are Question Marks: they have tiny revenue shares 3.9% and 5%, but are highly profitable (26.6% and 20.9% margins). Beverages, Dairy & Bakery, and Cleaning & Household are all Stars: they are both popular (above 10% revenue) and profitable(good margin around 14-15%). The good thing is there are no categories in Dog (low revenue and low margin).

The strategic imperative is to use the high traffic from the Staples Cash Cow to grow the sales of the high-margin Personal Care and Snacks Question Marks. This graph provides the core strategy: maintain Cash Cows, invest in Stars, and aggressively promote Question Marks.

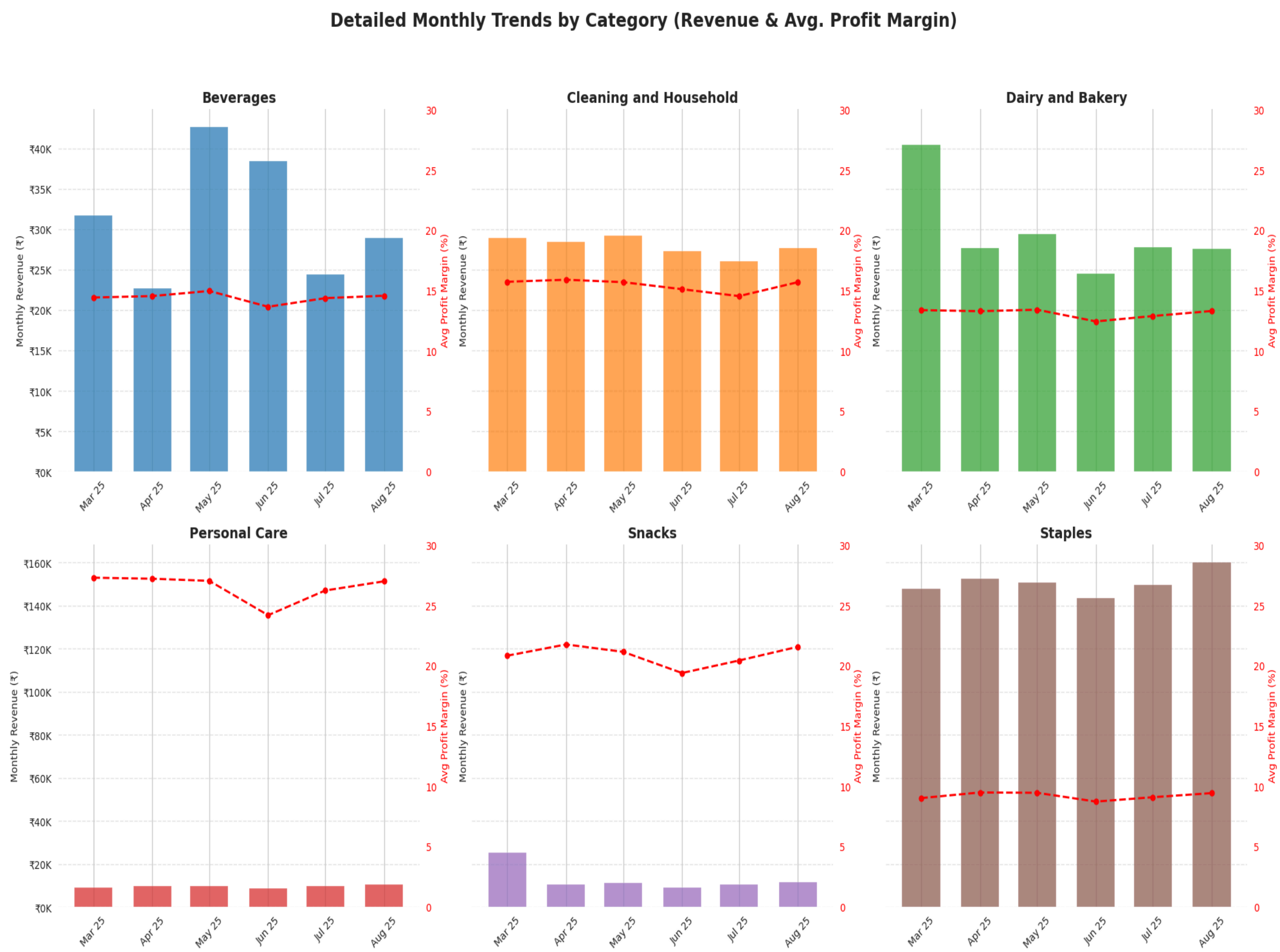

Figure 6: Detailed Monthly Trends by Category

This visual provides a detailed, 6-month performance breakdown for each of the six product categories. Each individual chart plots the Monthly Revenue (Rs) as bars (left axis) against the Average Profit Margin (%) as a line (right axis).

The Snacks category (avg. 20.9% margin) reveals a critical seasonal pattern. It had a massive revenue spike in March, reaching Rs 25,443, which was driven by the Ramzan festival as customers purchased items for evening celebrations. This revenue is more than double any other month. Following this festival period, sales normalized back to their baseline of ~Rs10k-Rs11k.

Stagnant "Question Marks": The other high-margin "Question Mark," Personal Care (avg.

26.6% margin), shows a completely flat and low revenue trend, proving it has no seasonal driver and is not growing on its own.

The "Cash Cow": Staples confirms its role with massive, stable revenue (consistently Rs 140k - Rs 160k) but a very low, stable margin (average 9.25%).

The "Stars": Cleaning & Household is a model of stability, with both its high revenue (~Rs 28k) and high margin (~15.5%) remaining consistent. Beverages showed strong seasonal potential with a clear revenue spike in May (Rs 42,722), which also correlated with its highest margin (15.01%).

The data reveals that the store's most profitable categories (Personal Care and Snacks) are underperforming. Personal Care sales are completely stagnant, while Snacks sales are highly seasonal, collapsing after the Ramzan festival.

3.3 Credit Management

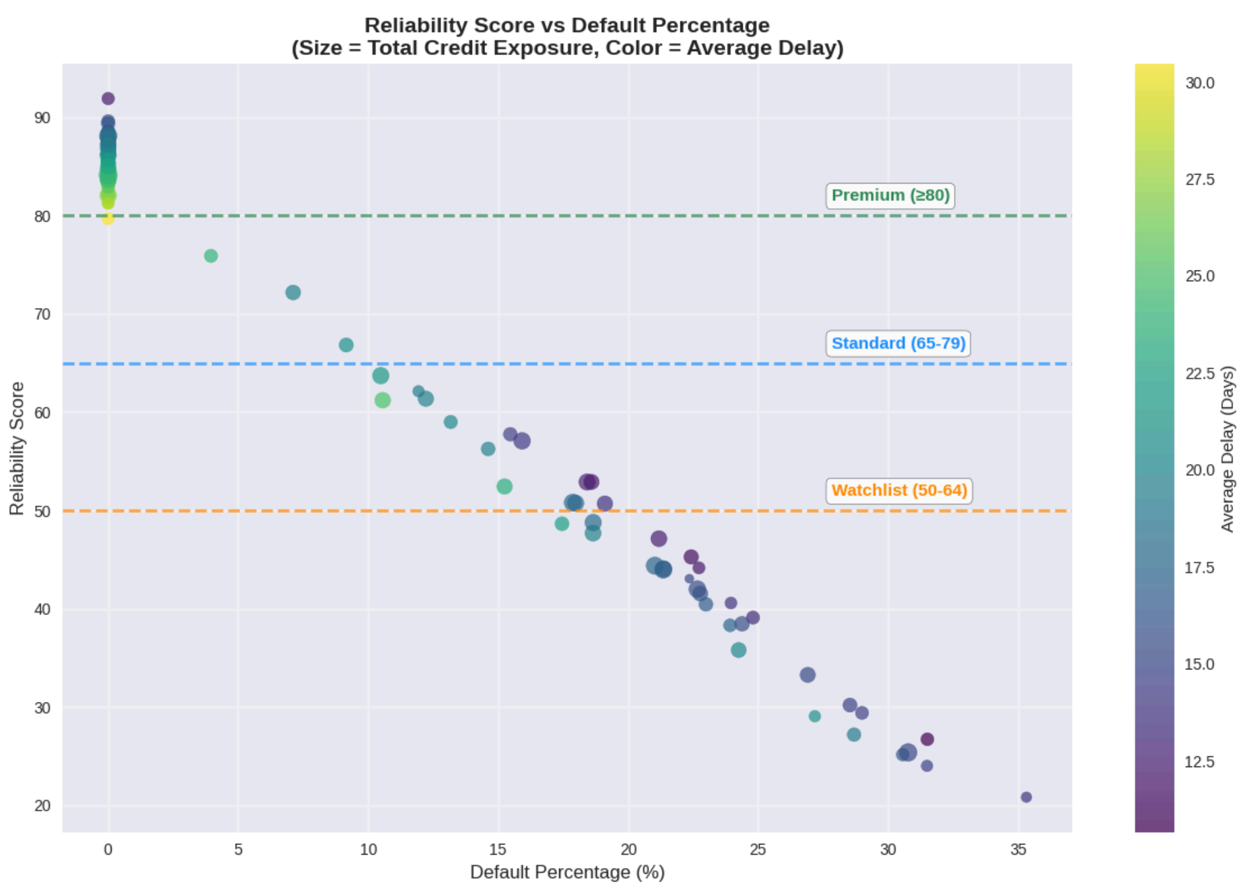

Figure 7: Reliability Score Model Validation Graph

This scatter plot validates the Reliability Score model by plotting customer reliability scores against actual default percentages. Bubble size represents total credit exposure, and color intensity indicates average payment delay days, offering a multi-dimensional risk view. This graph is necessary for proving the reliability of the model. The visualization reveals a clear inverse relationship—higher reliability scores consistently predict lower default rates. The model accurately segments all 105 customers into four risk tiers: Premium (≥80, 0% default), Standard (65–79, 3.9–9.1%), Watchlist (50–64, 10–20%), and Cash Only (<50, 20–35%).
It achieves 100% predictive accuracy in extreme categories, correctly identifying all zero-default Premium and high-default Cash Only customers (83% of the base),while the remaining 17% follow expected moderate risk patterns. . This validation confirms that the system enables data-driven, objective, and automated credit decisions, establishing a strong foundation for tier-based credit policies.

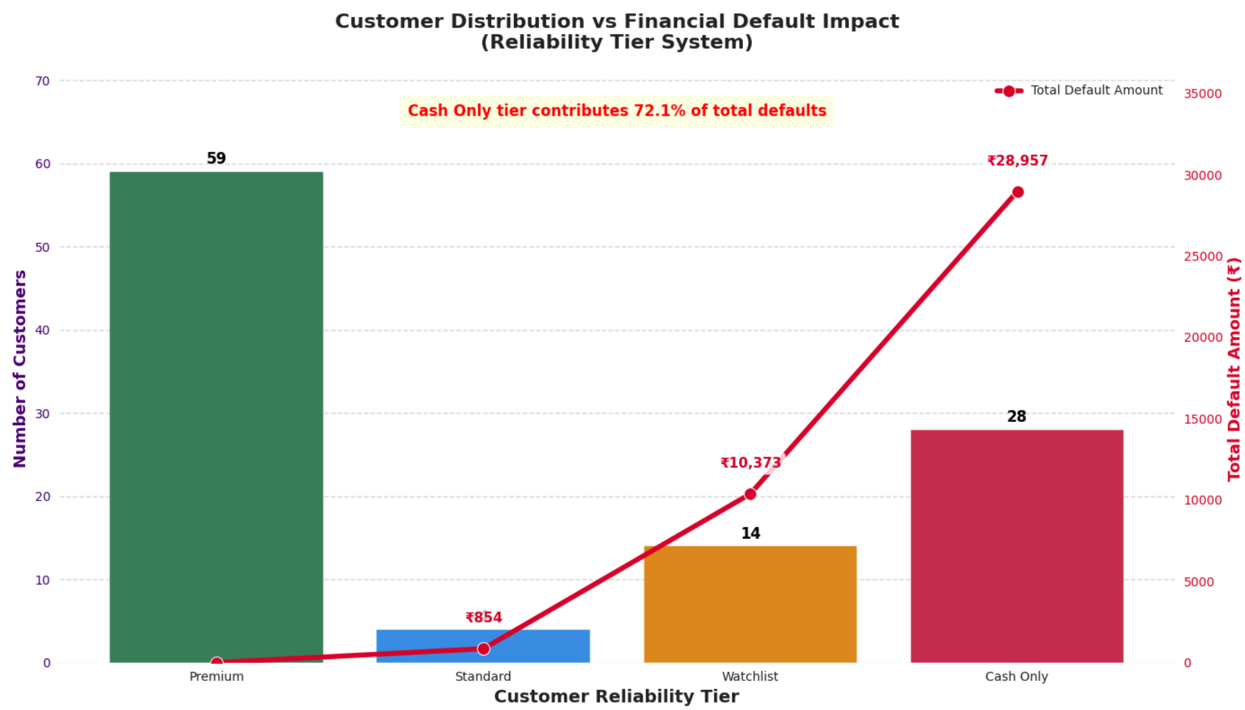

Figure 8: Customer Distribution vs Financial Default Impact

This dual-axis chart presents the core business problem uncovered by our new score. The bars (left axis) show the number of customers in each of the four reliability tiers. The red line (right axis) shows the total default amount in Rupees caused by each tier over the 5-month period.
The data reveals a severe and concentrated risk. The "Premium" tier (59 customers) is responsible for Rs 0 in defaults. The "Cash Only" tier (28 customers) is responsible for Rs 28,957 in defaults, 72.1% of total default, and the "Watchlist" tier (14 customers) is responsible for Rs 10,373, 25.9% of total default, Standard tier has only 4 customers with Rs 854 defaults, 2% of total default.
This is the key "problem" graph. It proves that 98% of all financial losses (Rs 39,330 of Rs 40,184) are caused by the "Watchlist" and "Cash Only" tiers. The problem is not widespread; it is specific and identifiable. This allows for a surgical solution that targets these two high-risk groups without penalizing our 59 best customers.

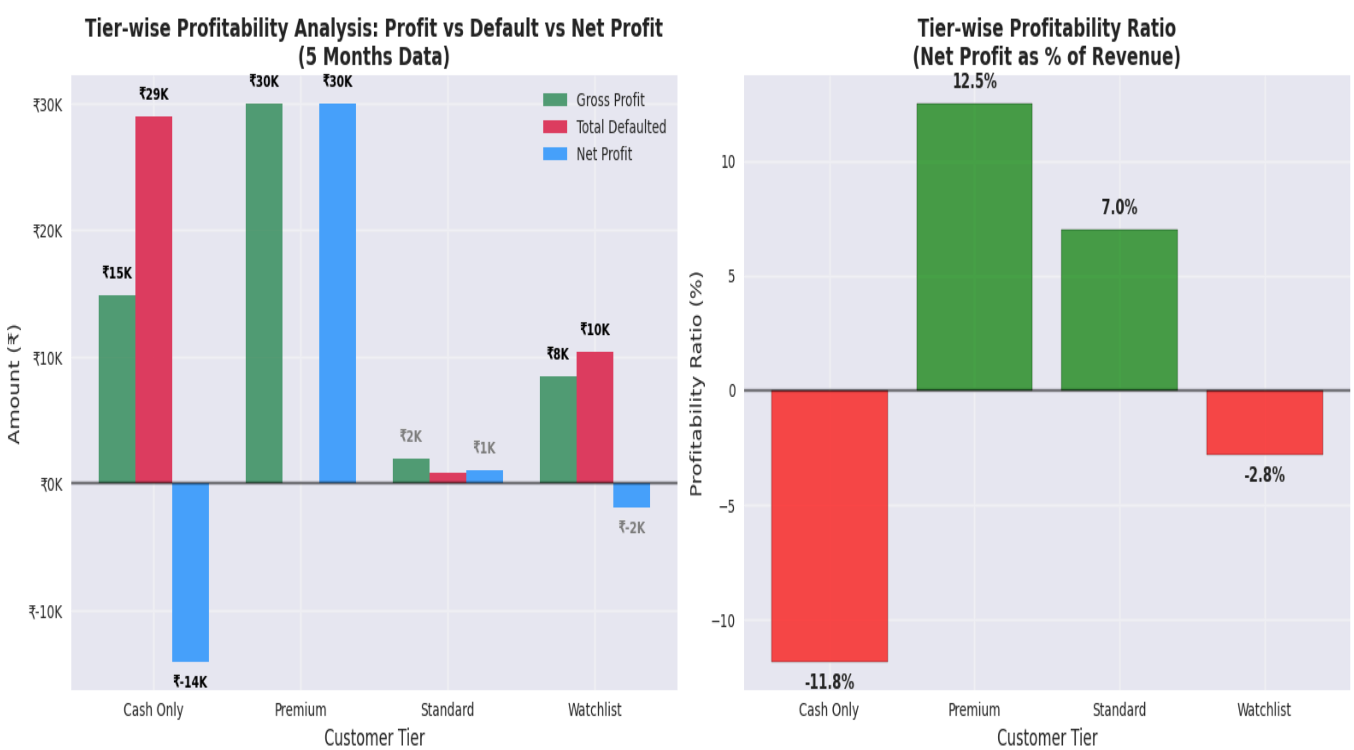

Figure 9: Tier-wise Profitability Analysis

This graph shows the actual profitability of each customer tier over the 5-month period. The left plot shows the Gross Profit (calculated from revenue), the Total Defaults, and the final Net Profit (in Rs). The right plot shows the final Net Profitability as a simple percentage.

The "Premium" tier is highly profitable, generating Rs 29k in Net Profit. The "Standard" tier is marginally profitable (Rs 1,090 Net Profit). The "Watchlist" tier loses Rs 2k. Critically, "Cash Only" tier is dangerously unprofitable., and loses a massive Rs 14k.

The second plot clearly tell the percentage of margin each tier contributed on their revenue, as you can see "Premium " tier has 12.5% net profit on revenue with 0% default while "Standard " tier contribute 7% net profit margin on their revenue, watchlist tier losing -2.8% on their revenue, but it is manageable with good policies, cash only tier burning the 11.8% on their revenue, which is highly dangerous.

It clearly shows that the shop is actively losing money on the "Cash Only " tier. Servicing this "Cash Only " on credit is destroying the profit generated by the good customers.

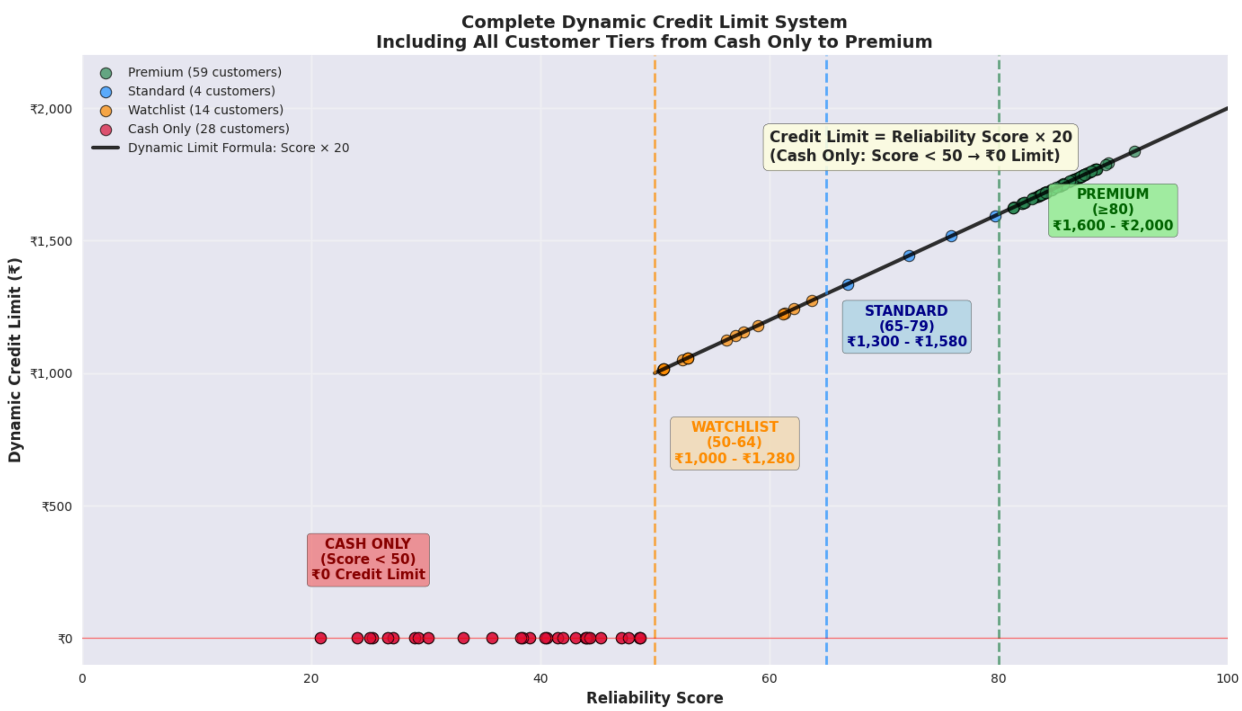

Figure 10: Complete Data-Driven Dynamic Credit Limit System

This scatter plot is the final, recommended solution. It visualizes the new Reliability Score × 20 policy for every customer. The X-axis shows their Reliability Score, and the Y-axis shows their new Recommended Credit Limit (Rs). The black line represents the formula, and each customer is colored by their new risk tier.

This policy is implemented in two simple, clear parts. First, all 28 customers in the 'Cash Only' tier (scores < 50) are moved to a Rs 0 limit. This immediately stops the 72% default (Rs 14,084 profit leak as identified in our Profitability Analysis) from this unprofitable group. Second, all 77 'Watchlist', 'Standard', and 'Premium' customers (scores >= 50) receive a fair, conservative credit limit that scales directly with their proven reliability, calculated as Limit = Reliability Score × 20.

This is the actionable policy that solves the problem. It is simple, data-driven, and directly fixes the profit leak identified in our previous analysis. It stops the losses from bad customers and continues to serve the good ones, perfectly balancing risk management with customer service.

4. Interpretation of Results and Recommendations

4.1 Interpretation and Significance of Findings

The analysis reveals that the business is caught in a "Volume vs. Value Trap"—it prioritizes high-revenue, low-margin categories (Staples, 57% revenue) while neglecting high-margin opportunities (Snacks, Personal Care). This strategic misalignment, combined with a deteriorating cash-to-credit ratio (from 3.42 to 1.86) and a concentrated credit risk (72% of defaults from 26.7% of customers), is systematically eroding profitability. The shop's 12.5% average margin, which is below the industry standard and 9.45% net profit margin(3.05% goes in default), is a direct result of this operational inefficiency. These findings identify not just symptoms, but the root causes of weak financial health, enabling targeted, high-impact interventions.

4.2 Actionable SMART Recommendations

Critical Priority: Stop Financial Bleeding (Implementation: Immediate – 1 Month)

1. Eliminate Credit for High-Risk Customers

    - **Specific:** Move all 28 "Cash Only" tier customers (Reliability Score < 50) to a Rs 0 credit limit.
    - **Measurable:** Stop 100% of the Rs 28,957 in defaults from this tier.
    - **Achievable:** A simple policy change enforced at the point of sale.
    - **Relevant:** Directly addresses 72% of the total credit default problem.
    - **Time-bound:** Implement within the next 30 days.

2. Launch Targeted Weekend Promotions

    - **Specific:** Dedicate Saturday and Sunday promotions to Snacks and Personal Care categories.
    - **Measurable:** Aim to increase revenue from these categories by 25% on weekends.
    - **Achievable:** Use existing weekend customer traffic and shelf space.
    - **Relevant:** Leverages peak traffic to solve the low-margin problem.
    - **Time-bound:** Design promotions within 2 weeks; launch and run for 3 months.

3. Manage Policy Transition with 1-Month "Data-Driven" Discount

    - **Specific:** Launch a store-wide, 1-month-only "2% Cash Discount" for *all* customers, positioned as a celebration of the new, fair, "data-driven" credit system.
    - **Measurable:** Retain at least 80% of the "Cash Only" tier customers (preventing churn) and increase overall cash sales during the 30-day period.
    - **Achievable:** This is a simple, temporary promotional change. The 2% cost (approx. ₹5,200) is a minor one-time investment to protect a major revenue stream.
    - **Relevant:** This action directly mitigates the business risk of customer churn, softens the transition for the "Cash Only" tier, and rewards loyal cash-paying customers.

- **Time-bound:** This 2% discount will run *only* for the first 30 days of the new policy implementation.

Strategic Priority: Drive Profitability (Implementation: Short-Term – 1–3 Months)

4. Implement the Dynamic Credit Limit System
    - **Specific:** Roll out the "Reliability Score × 20" max credit limit formula for all 77 customers with a score ≥ 50, while moving all 28 "Cash Only" customers (score < 50) to Rs 0 credit limit.
    - **Measurable:** Reduce overall default rates by 72% immediately by eliminating Cash Only tier defaults, with a goal to reach 85% reduction by addressing Watchlist and Standard customers through payment reminders.
    - **Achievable:** The scoring model is validated and customers are clearly identified; requires systematic application and customer communication.
    - **Relevant:** Directly addresses the root cause of credit losses while maintaining service for reliable customers.
    - **Time-bound:** Full implementation and communication to customers within 90 days.

5. Execute a Product Portfolio Shift
    - **Specific:** Use Staples as a loss-leader by creating bundled offers with high-margin Snacks and Personal Care items. Implement strategic shelf placement and weekend promotions for high-margin categories.
    - **Measurable:** Increase the gross profit margin by 1.5% (from 12.5% to 14%) within 6 months.
    - **Achievable:** Requires supplier negotiation and in-store marketing, but no new inventory.
    - **Relevant:** Directly addresses the issue of being below the industry margin.
    - **Time-bound:** Develop bundles in Month 2; monitor and adjust for 4 months.

Sustainability Priority: Build Long-Term Health (Implementation: Medium-Term – 3–6 Months)

6. Formalize a Tier-Based Customer Loyalty Program
    - **Specific:** Offer small discounts and payment flexibility for "Premium" and "Standard" tier customers.
    - **Measurable:** Increase the proportion of "Premium" customers by 10% in 6 months.
    - **Achievable:** Low-cost rewards that incentivize good payment behavior.
    - **Relevant:** Rewards reliable customers and encourages others to improve their credit habits.
    - **Time-bound:** Design the program in Quarter 2; launch by the start of Quarter 3.

4.3 Projected Impact

Implementing these recommendations is expected to create a powerful compound effect. The immediate application of the Reliability Score–based credit system is expected to eliminate 72.1% of all defaults (from the "Cash Only" tier), with a clear path to an 85% total default reduction, saving approximately Rs 34,000 in losses.

Simultaneously, the "Product Portfolio Shift" strategy—focusing on high-margin categories such as Snacks and Personal Care—is projected to achieve the targeted 1.5% margin improvement, contributing an additional Rs 19,600 in profit.

The transition will be supported by a one-month 2% store-wide discount campaign, positioned as a celebration of the new data-driven system launch, ensuring customer retention during this strategic shift.

This holistic approach addresses cash flow, profitability, and risk simultaneously, leading to a combined net profit increase of over Rs 53,700. This represents a 43.4% profit growth—transforming the business from a struggling 9.45% net margin operation to a professionally managed 13.5% net margin enterprise. This demonstrates a sustainable, data-driven, and high-confidence improvement in the shop's overall financial health.

5. Acknowledgements

I would like to express my sincere gratitude to my professors, teaching assistants, and the entire BDM capstone project team at IIT Madras for their invaluable guidance and support throughout this project. Their insights and feedback were instrumental in shaping this analysis and helping me grow as a data professional.

I am particularly grateful to Dr. Aditya Chandel and Dr. Ashwin J. Baliga for preparing comprehensive documentation, organizing the project bootcamp that cleared all our doubts, and providing continuous mentorship throughout this learning journey.

Special thanks to my TA mentor, who provided exceptional clarity during the weekly bootcamp sessions by patiently addressing my queries and guiding me through the entire BDM project lifecycle.

I also extend my appreciation to the owner of Faiz General Store for providing access to business data and valuable operational insights that made this real-world analysis possible.

---

6. Project Links and Datasets

- Colab Notebook (Analysis Link): **IITM_BDM_Project_Colab_Notebook.ipynb**
- Google Sheet (Dataset Link): Faiz General Store -Data